KenduriFlow System Flows

Oct 23, 2026 · @waz

KenduriFlow turns one host registration into a parking plan, an invitation link with entry and exit QR passes, a live gate screen for the guard and a township dashboard. This doc walks through every flow in the order a majlis happens, with the real numbers and rules from the working system.

The problem and the answer

A 500-guest wedding brings about 167 cars into a street whose kerb holds 26. Guests park on both sides, block gates, and close the road an ambulance needs. Councils already receive hundreds of complaints about cars blocking homes and roads: Pasir Gudang alone recorded 349 in the first nine months of 2023 (Malay Mail).

KenduriFlow solves it in four steps:

1. **Predict.** Cars = guests ÷ people per car, spread over the usual arrival pattern for that type of event. Plan for the busiest moment + 10%.
2. **Allocate.** Fill spaces nearest first: the host's own street, neighbours' shared driveways, the next street, the surau, then the mall with a shuttle. The house side of the road is never used.
3. **Guide live.** Each guest holds the next free space on the way and passes the gate with an entry QR; the closest spaces are kept for elderly and OKU guests; every screen updates in real time.
4. **Learn.** The guard's real counts are saved when the event ends and tune the next estimate for that type of event.

It runs in any phone browser and needs no new hardware: the guard uses their own phone as the counter and QR scanner.

Who uses it

Four kinds of people use KenduriFlow, each with their own screens. Residents, guards and the township team sign in; guests never need an account.

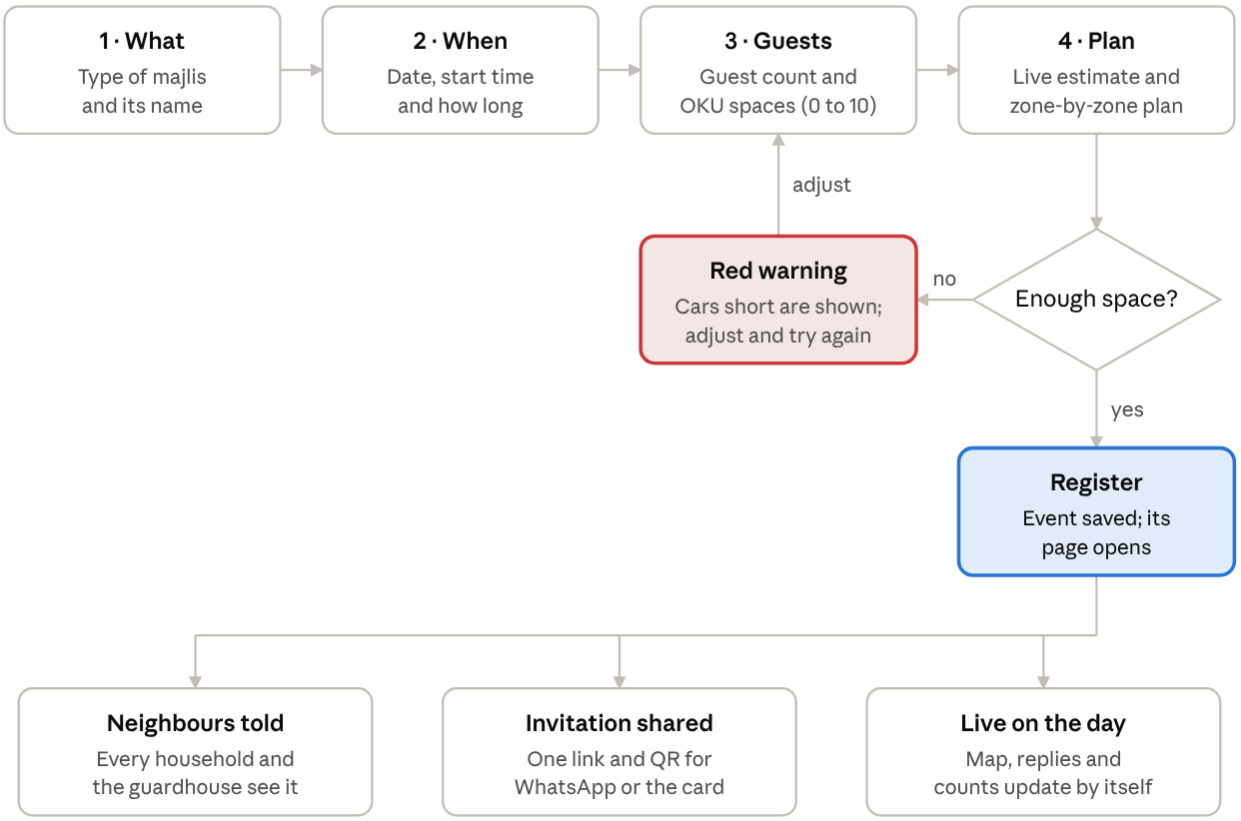
Role	Main screens	What they do
Resident (host)	My events, Plan a majlis wizard, event page	Registers a majlis in 4 steps, gets the parking plan, shares one invitation link, watches it live
Resident (neighbour)	Home	Gets early notice of nearby events, shares a driveway for 20 points per car, keeps the kerb opposite their gate clear, reports a blocked gate in one tap
Guest	Invitation link (no app, no account)	Replies with people and cars, sees where to park and spaces left, holds a space on the way, shows an entry QR, then an exit QR
Guard	Gate screen	Scans entry and exit QR passes, counts cars in and out, sends elderly or OKU cars to the nearest spaces, recounts zones, handles the emergency lane and reports, ends the event
Township team	Dashboard, People	Sees every event, street clashes, reports and lane alerts, what the counts taught the system, planning figures; adds residents and guards

The guest page and neighbour home work in English and Bahasa Melayu.

Flow 1: Host plans a majlis

The host answers four short questions and sees the parking plan before registering; one tap then tells the neighbourhood and gives them the invitation link.

One registration gives a plan, a guest link and notice for neighbours



host flow · 4 wizard steps, 1 check, 3 results

A resident always hosts at their own house, so the address is filled in. The same house cannot be booked twice for overlapping times. After registering, a sheet offers **Share invitation**, which opens the phone's share menu (WhatsApp, Telegram, SMS) with a ready message and the link. The event page then has four tabs: Overview (cars through the day, plan), Live (map and zones), Guests (link, QR, replies) and Neighbours (shared driveways, reports, timeline).

How the prediction works

The 500-guest wedding needs 167 cars in total but only 70 at the same time, at 1:40 pm, so the plan is 77 spaces, not 167.

1. **Total cars** = guests ÷ people per car, rounded up: 500 ÷ 3.0 = 167.
2. **Arrivals** are spread across the event in 5-minute steps along a bell curve centred on the usual busy time for that type of event.
3. **Cars at the house** at any moment = cars that arrived within their usual stay.
4. **Spaces to plan** = the busiest moment + 10%: 70 × 1.1 = 77.

Event type	People per car	Usual stay (min)	Arrival pattern
Wedding	3.0	60	Waves, busiest about a third of the way in
Kenduri doa selamat	2.8	75	Most in the first hour for the doa
Tahlil	1.6	60	Almost everyone at once, after Maghrib
Aqiqah	2.8	60	Early, for the doa and the meal
Open house	3.5	45	Small groups spread across the day
Birthday	2.5	120	At the start, staying to the end

Two things sharpen the starting numbers:

- **Guests' replies.** Before the day, guests answer "how many people, how many cars" from their link. After 5 replies, their people-per-car gradually takes over, with weight = replies ÷ (replies + 15). The guest count never drops below the people who replied. In the sample walimah, 42 replies at 2.5 people per car raised the estimate from 117 to 134 cars.
- **Past counts.** Each type's people-per-car blends the starting value (counted as 4 events) with real counts from past events (see Flow 6).

While the host fills in the wizard, the estimate and plan update live, so they see the effect of every change.

How the parking plan is allocated

The 77 spaces fill nearest first: 55 cars park within walking distance and 22 go to the mall, served by 2 shuttle buggies every 6 minutes.

Zone	Spaces planned	Available to this event	Walk to the house
Street kerb, Jalan Elmina 3/2 (own street)	16	16 (the other 10 go to an aqiqah at No. 20 the same afternoon)	1 min
Shared driveways (6 neighbours)	11	11	1 to 2 min

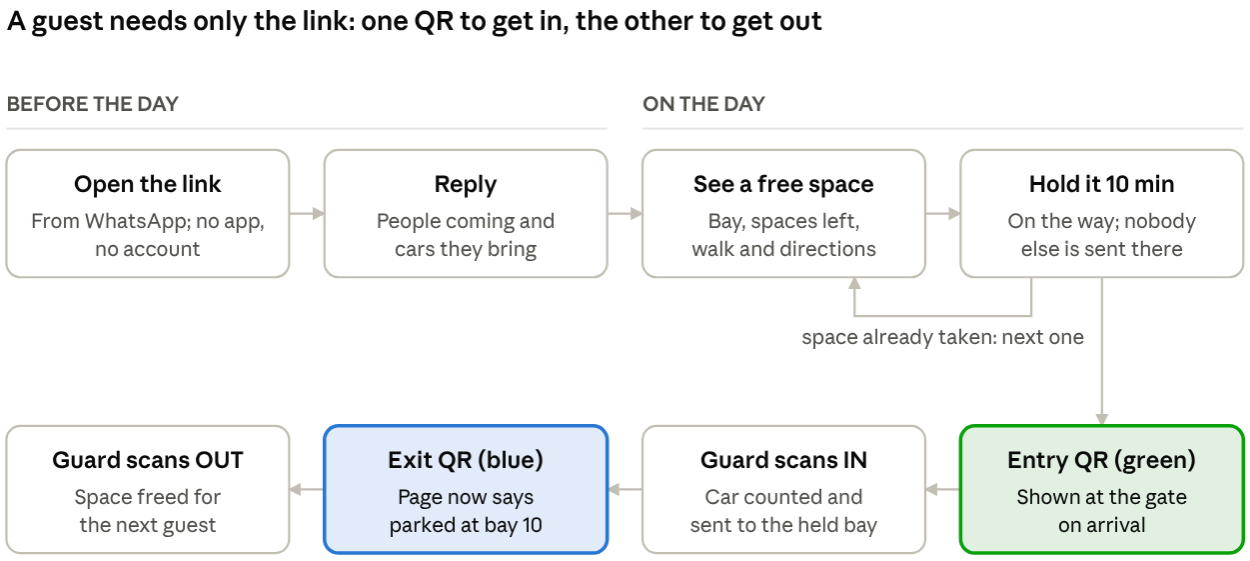
Zone	Spaces planned	Available to this event	Walk to the house
Street kerb, Jalan Elmina 3/3 (next street)	16	16	3 min through the linear park
Surau Al-Ikhlas car park	12	12	4 min
Lakeside Mall car park, level P2	22	80	Shuttle, 6 min ride
Total	**77**		

The rules behind it:

- Only the park side of each street is used. The house side stays clear as the emergency lane.
- The kerb spaces nearest the house are kept for elderly and OKU guests (the host chooses 0 to 10).
- Driveways are used only when the owner has shared them for this event, on the same street.
- Bays opposite a neighbour's gate are skipped when that neighbour asked to keep them clear.
- Two events at the same time on one street: each kerb space goes to the host whose house is nearer.
- Shared car parks (surau, mall): the event registered first keeps the spaces it planned.
- The surau car park is not used during Friday prayers (12:15 pm to 2:30 pm).
- Mall overflow gets a shuttle: one 8-seat buggy per 40 riders, running every 12 minutes divided by the number of buggies (at least every 3 minutes).
- Not enough space anywhere: the host sees a red warning with the number of cars short before registering.

Flow 2: Guest journey

A guest needs nothing but the invitation link: it gives them a reply form, a free space held for them, a green entry QR and, once inside, a blue exit QR.



guest journey · 8 steps, before the day and on the day

What the guest page shows and does:

- **Where to park:** the exact bay (e.g. Kerb 3/2, bay 10), spaces left there and in total, walk time or shuttle, directions, and a map with the walking route to the house.
- **Elderly or OKU switch:** sends the car to the spaces kept next to the house.
- **Someone is already parked here:** marks that bay full and shows, or holds, the next one.
- **Live:** the page updates by itself as spaces fill; on weak signal it keeps the last plan and says how old it is.
- **Two backup codes:** each QR has a 6-letter code under it for the guard to type if the camera fails.
- **Languages:** English or Bahasa Melayu, one tap at the top.

Flow 3: Guard at the gate

The guard's own phone is the counter and the QR scanner: every car is recorded in and out, so the free-space count every guest sees is right.

The guard records every car in and out, with or without a QR pass



guard at the gate · in and out, with or without a pass

The Gate screen has four tabs (Count, Map, Reports, Activity) and a **Scan** button on every tab. Its top tiles show cars here now, cars arrived against the prediction, guests on the way (spaces held) and spaces free, with the next bay to send a car to. The camera scanner reads QR codes on any phone and keeps scanning car after car, with a torch for dark gates; scanning the same entry QR twice says "Already in" and counts nothing. When the event ends, the guard taps **End event and save counts**; if they forget, it ends by itself 2 hours after the end time.

Flow 4: Neighbours

A neighbour learns about every nearby majlis as soon as it is registered, and can help or complain in one tap.

1. **Early notice.** The host registers; every other household in the precinct sees the event on their home screen, marked New, Today or Tomorrow, with how many cars to expect and where they will park.

2. **Share a driveway.** Going out that day? The neighbour switches on "Share your driveway" for 1 or 2 cars (same street only). Guests can now be sent there, and the neighbour earns 20 points per guest car.
3. **Keep my gate clear.** For a neighbour who must be able to drive out (shift work, someone at home who may need a car quickly), the kerb bays opposite their driveway are left out of every event's plan. In the demo, No. 21 has it on.
4. **Report a problem in one tap.** "My gate is blocked" goes straight to the guard for the event most likely causing it. Three more issues sit one tap further: cars in the emergency lane, a car on the grass verge, a crowd outside the gate.
5. **Follow up.** The report shows on the guard's Reports tab (with a red count) and the township dashboard; the neighbour sees "Sent to guard", then "Sorted by guard".

Flow 5: Emergency lane alert

The house side of the road must stay open for ambulances and fire engines; one tap from the guard or a neighbour turns every screen red until the guard clears it.

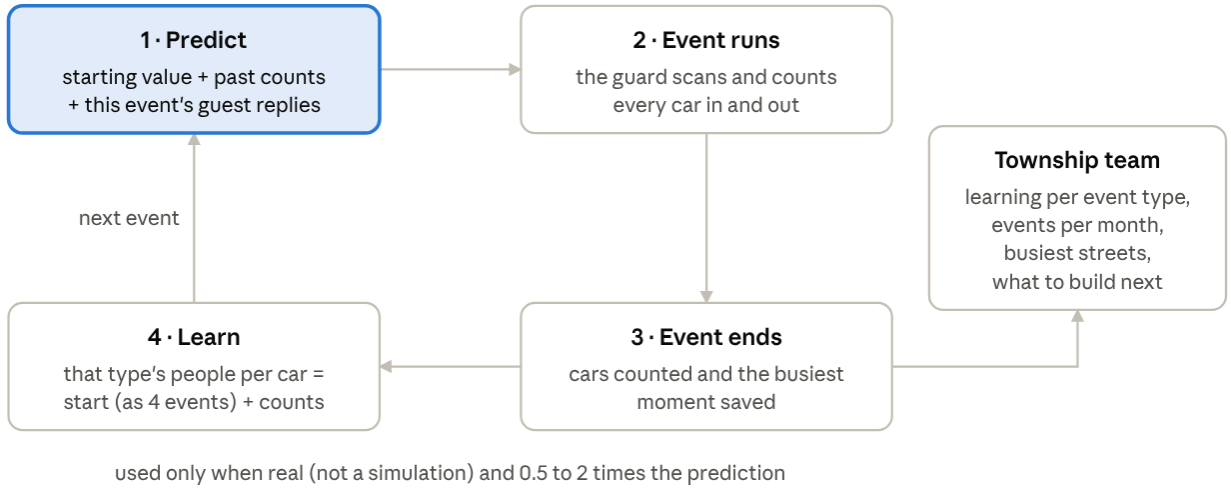
1. **Someone sees a car in the lane.** The guard taps **Car in lane** on the Gate screen, or a neighbour taps **Cars in the emergency lane** on their home screen.
2. **Every screen shows it.** The guard's lane banner turns red and the phone buzzes; the live map tints the house side of that street red; guests on that street see "Never stop on the house side of the road"; the township dashboard counts a lane alert.
3. **The guard acts.** They ask the driver to move to the next free space the app shows.
4. **Lane clear.** The guard taps **Lane clear**. It removes every car recorded in the lane on that street and closes the neighbours' reports about it; all screens turn green again.

There is no camera on the road: the alert comes from people who see the car, which needs no new hardware.

Flow 6: Township team and the learning loop

Every event that ends with real counts tunes the next estimate for that type of event, and gives the township team the figures to plan with.

Every finished event makes the next estimate more accurate



learning loop · 4 steps, feeding the township dashboard

The blend for each event type:

$$\text{people per car} = \frac{\text{starting value} \times 4 + \sum \text{counted values}}{4 + \text{counted events}}$$

In the demo's past events, three counted weddings averaged exactly 3.0 people per car, so the wedding estimate is confirmed (still 167 cars); two kenduri moved it from 2.8 to 2.72, and three tahlil from 1.6 to 1.51. An event never tunes its own estimate, so a plan stays stable while it runs.

The township dashboard has five tabs' worth of views: upcoming events with cars at the busiest time, walking-distance and shuttle split; street clashes and extra rubbish pickups (booked for 8:00 am the day after each event); neighbour reports and lane alerts; the learning table per event type; and planning figures (events per month, busiest streets, suggestions such as visitor bays where a street's average peak exceeds its kerb). The People page is where the team adds residents and guards.

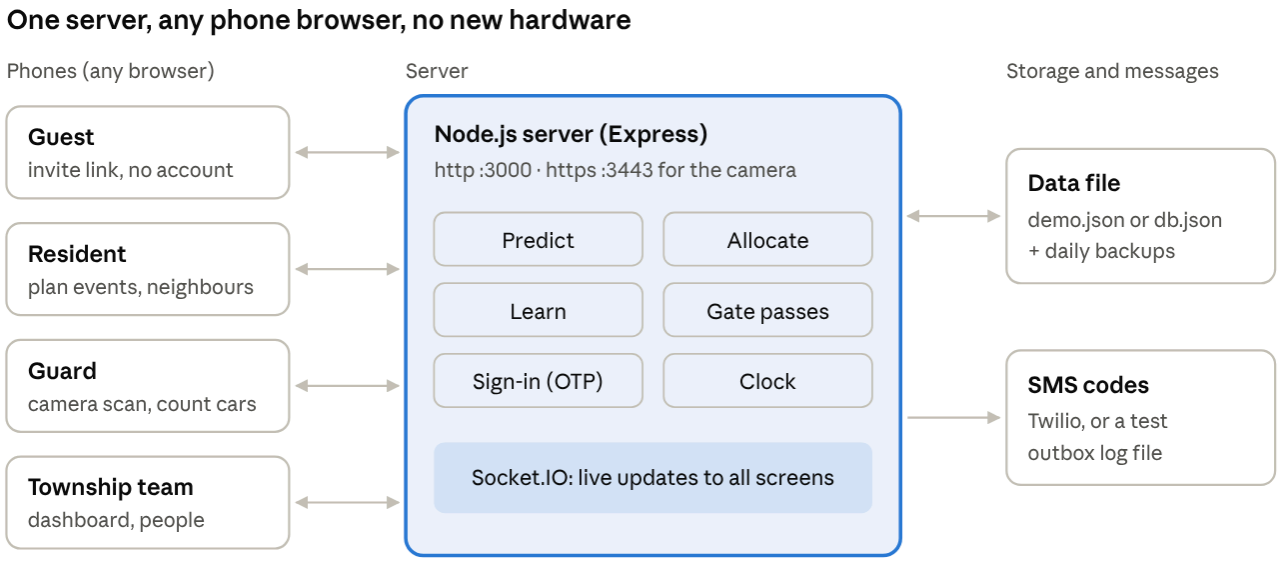
Real-life cases it handles

Each of these is built in and was tested on the working system.

Situation	What KenduriFlow does
Two events on one street at the same time	Each kerb space goes to the nearer host; both hosts see the clash; shared car parks go to the event registered first
Friday prayers overlap the event	The surau car park is left out of the plan
Not enough space for the guests	Red warning with the number of cars short, before the host registers
Same house booked twice for overlapping times	Refused, naming the event already booked
Two guests sent to the same space	Prevented: a guest on the way holds the space for 10 minutes; nobody else is sent there
A stranger is parked in the space shown	The guest taps "Someone is already parked here"; it is marked full and the next space appears
Entry QR scanned twice	"Already in": nothing is counted twice
Exit QR for a car that never came in	"Not inside": nothing to free
Elderly or OKU passenger	Guest switches it on (or the guard taps the OKU button); the car goes to the spaces kept next to the house
Camera fails at the gate	The guard types the 6-letter code under the QR
Weak signal at the street	The guest page keeps showing the last plan, with how many minutes old it is
Guard forgets to end the event	It ends by itself 2 hours after the end time and saves the counts
A neighbour must always be able to drive out	Keep my gate clear: the bays opposite their gate are never used
A car blocks the emergency lane	Flow 5: red on every screen until the guard marks it clear
Guard's count drifts	Recount: the guard enters the cars they can see in a zone

Architecture

Everything runs on one small Node.js server. Every role uses a phone browser, so nobody installs anything from an app store, and the township buys no sensors or cameras.



Installs as a phone app (PWA) and keeps working on weak signal. The guard uses their own phone camera.

architecture · 4 phone roles, 1 server, 1 data file

The server keeps all data in one JSON file, written safely (a temporary file, then a rename) and copied to a daily backup. The demo uses its own file, so practice never touches real data. When anything changes (a car scanned in, a space taken, a lane report), Socket.IO pushes the new state to every open screen within a second. Local https on port 3443 exists only because phone browsers allow the camera on secure pages; in production the system sits behind a normal https proxy. A Dockerfile is included for deployment.

Demo and real system

The same code runs two ways: a demo for judges with no sign-in, and a real system with proper accounts. They never share data.

	Demo	Real system
Start	`npm start`	`npm run live`
Getting in	Choose a role: Township team, Guard, Resident or Guest	Mobile number + 6-digit code by SMS
People	Four demo people	Added by the township team on the People page

	Demo	Real system
Data	Sample events (`data/demo.json`)	Real events (`data/db.json`), a copy kept every day for 14 days
Clock	"Play the event" fast-forwards a whole event in under a minute	Real time in Malaysia
Dashboard figures	Sample figures	Counted from real events

How the real system protects people:

- **First start:** the first township account needs a one-time setup key printed only in the server terminal.
- **Sign-in codes:** 6 random digits, valid 5 minutes, 5 wrong tries, at most 3 codes per number in 10 minutes; stored only as a scrambled hash. The page answers the same way for unknown numbers, so it cannot reveal who lives where.
- **Permissions:** every action is checked on the server against the person's role and house. A resident can only manage their own events and report from their own house; only guards and the township team can count cars.
- **Privacy:** event details need sign-in; a guest sees only their own invitation page, without neighbours' reports or the guard's log; invitation links, passes and car records use random, unguessable ids.
- **Web safety:** changes must be sent as JSON (blocks other websites acting for a signed-in person), strict security headers, and rate limits on guest buttons.
- **Removing a person** signs them out at once; their past records stay.

Suggested slide outline

Twelve slides, one idea each; the diagrams in this doc can go straight onto slides 4 to 10.

1. **Title.** KenduriFlow: smart parking for every township celebration.
2. **The problem.** 500 guests, 167 cars, 26 kerb spaces; blocked gates and emergency lanes; 349 complaints in Pasir Gudang in nine months of 2023.
3. **The answer in four steps.** Predict, allocate, guide live, learn.
4. **Who uses it.** Host, guest, guard, neighbour, township team (roles table).
5. **Host flow.** Plan a majlis in 4 steps, register, share one link (Flow 1 diagram).
6. **The prediction.** 167 cars but 70 at once, plan 77 (prediction section).
7. **The parking plan.** Nearest first, 55 within walking distance, 22 by shuttle (allocation table).

8. **Guest flow.** Invitation link, hold a space, green entry QR, blue exit QR (Flow 2 diagram).
9. **Guard flow.** Scan in and out, count, OKU, recount, end the event (Flow 3 diagram).
10. **Neighbours and the emergency lane.** Early notice, driveway points, one-tap report, lane alert (Flows 4 and 5).
11. **It learns.** Real counts tune the next estimate; township planning data (Flow 6 diagram).
12. **Built and ready.** Any phone browser, no new hardware, demo and real system, what comes next (architecture diagram, cases table).

Live demo order for judges: choose **Resident** and open the wedding; choose **Guest** on a phone and show the entry QR; choose **Guard** and scan it; choose **Township team** to show the dashboard.